

Simpson's Paradox



The Setup: Choosing a Surgeon

- Imagine you need surgery and compare two doctors' track records:
 - **Dr. Jones:** 86% overall success rate
 - **Dr. Smith:** 79% overall success rate
- The numbers seem to make the choice for you

“ **Question:** *Is picking the doctor with the higher success rate always the rational choice?* ”

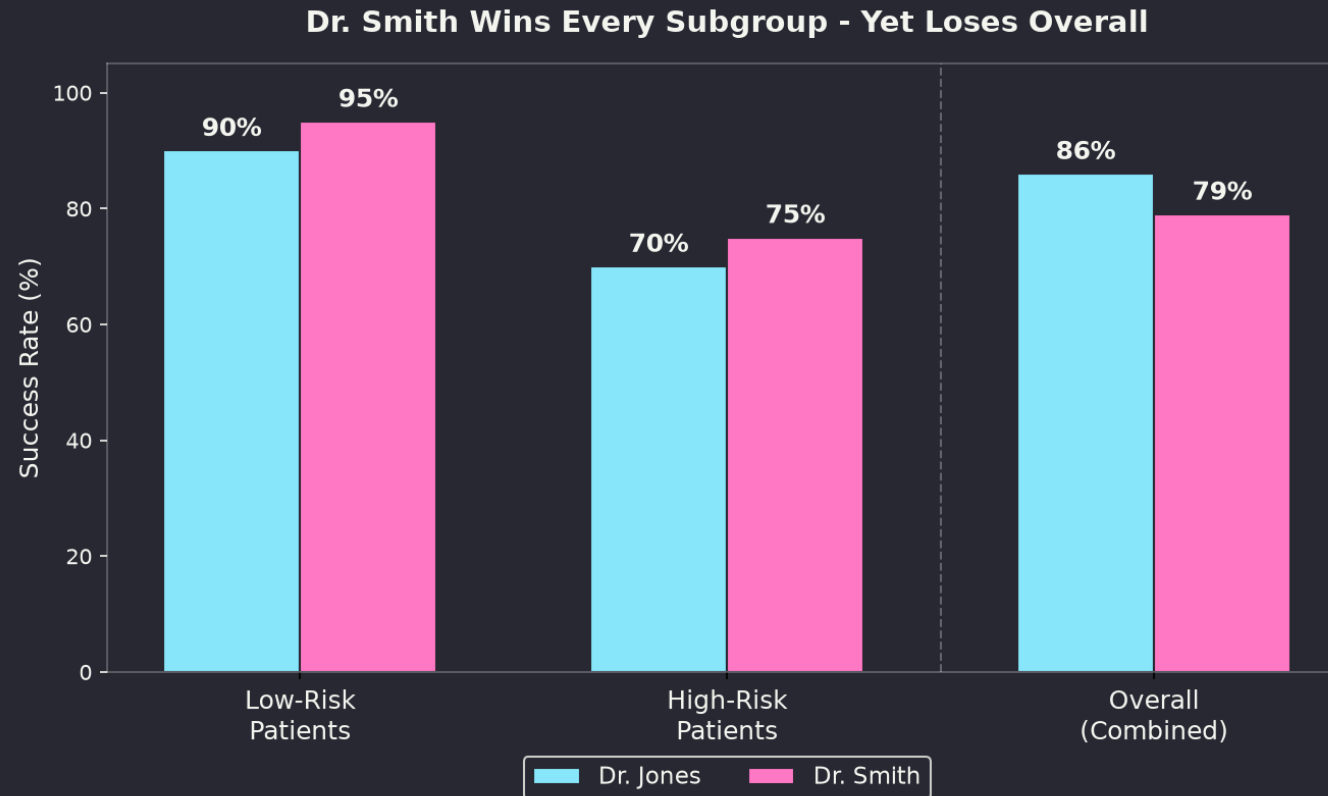
Watch: Simpson's Paradox Explained

- The surgeon example and worked math in this deck are drawn from this video
- [Simpson's Paradox: Why More Data Can Trick You](#)

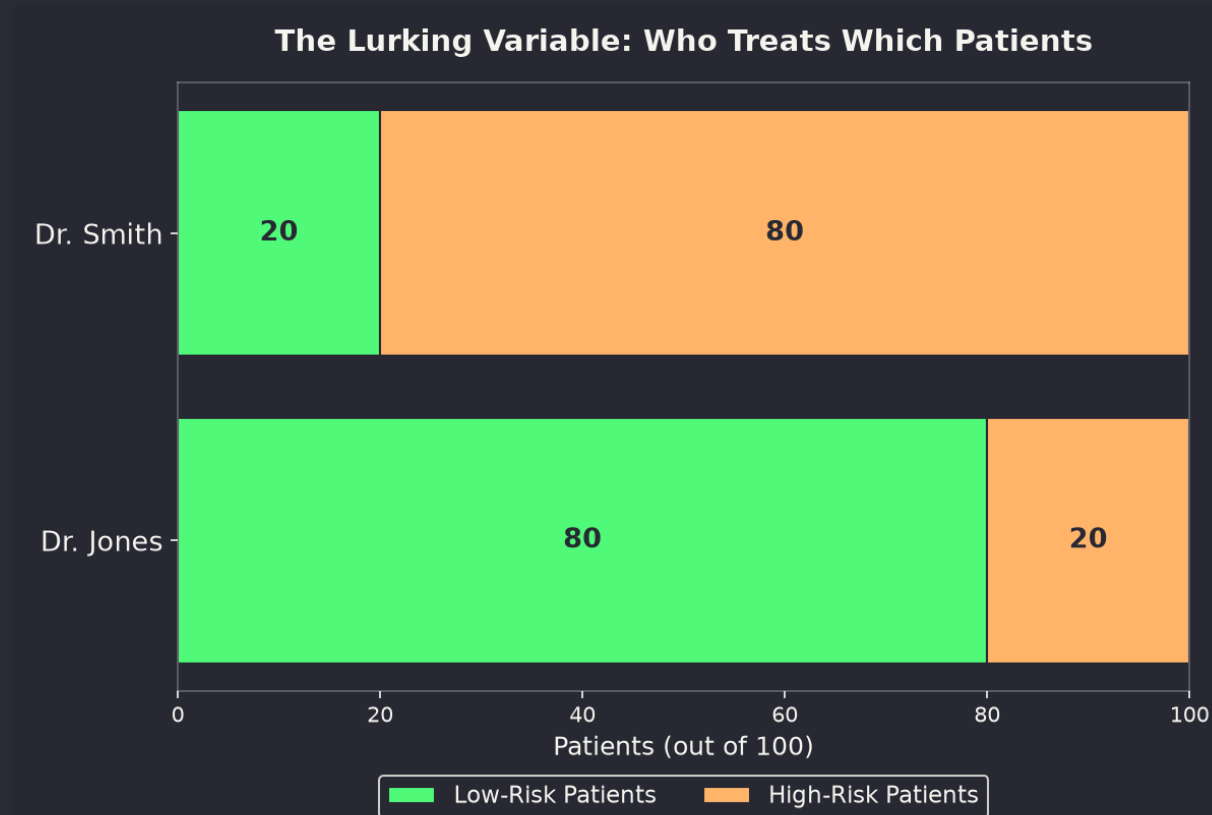
Digging Deeper: Split by Patient Risk

- Ask the hospital for outcomes broken down by patient type
- **High-risk patients:**
 - Dr. Smith: **75%** success
 - Dr. Jones: **70%** success
- **Low-risk patients:**
 - Dr. Smith: **95%** success
 - Dr. Jones: **90%** success
- Dr. Smith wins **both** subgroups — yet loses overall

The Numbers, Visualized



How Is This Possible? The Caseload Mix



Doing the Math: Weighted Averages

Dr. Smith (20 low-risk, 80 high-risk)

- Low-risk: $20 \times 95\% = 19$ survivors
- High-risk: $80 \times 75\% = 60$ survivors
- Total: $79 / 100 = 79\%$

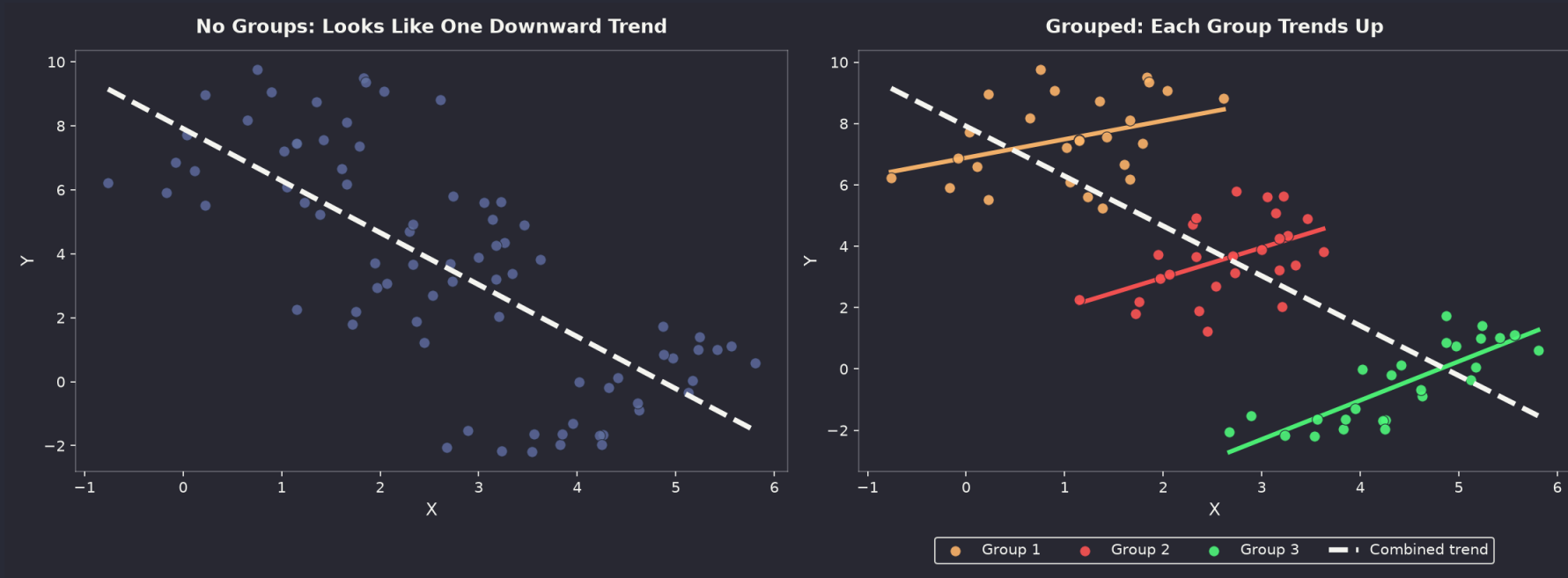
Dr. Jones (80 low-risk, 20 high-risk)

- Low-risk: $80 \times 90\% = 72$ survivors
- High-risk: $20 \times 70\% = 14$ survivors
- Total: $86 / 100 = 86\%$

Naming the Phenomenon

- **Simpson's Paradox:**
 - a trend appears in several groups of data
 - but disappears or reverses when the groups are combined
- Variables can be **positively correlated** within every subgroup
 - yet **negatively correlated** in aggregate
- The overall number told you to pick Jones
- The subgroup breakdown says Smith is the better choice,
 - **for either kind of patient**

Trend Reversal, Visualized



- **Left:** no group labels — looks like one clean downward trend
- **Right:** same data, colored by group — each group actually trends **upward**
- No single point moved — only whether the grouping was visible

The Lurking Variable

- A **lurking (confounding) variable** is a hidden factor
 - that changes what the data appears to say
- In this example:
 - it is the **proportion of high-risk vs. low-risk patients** each surgeon treats
 - It's correlated with:
 - The **outcome** (high-risk patients survive less often, regardless of surgeon)
 - The **group** being compared (Smith sees far more high-risk cases)
 - Ignore it, and the aggregate comparison is comparing apples to oranges

Real-World Case: Hospital Rankings

- Major referral centers and trauma hospitals
 - often look **worse** on raw survival-rate rankings
- Reason:
 - they take the sickest, most complex patients that smaller hospitals decline to treat
- Once outcomes are **risk-adjusted** for patient severity,
 - top referral centers frequently rank at the top
- Raw league tables can punish the hospitals doing the hardest, most valuable work

“ *Source: Krumholz et al. (2006), risk-standardized mortality methodology — see References* ”

Real-World Case: Clinical Trials

- A new treatment can look **weaker overall** than a control or existing treatment
- But the same treatment can **outperform**
 - once you stratify by disease severity or baseline risk
- Cause:
 - sicker patients may be disproportionately assigned to (or seek out) the new treatment
- Trial design and analysis must account for this
 - via **stratification** or **randomization checks**

“ *Classic case: Charig et al. (1986) kidney stone treatment trial — see References* ”

Real-World Case: UC Berkeley Admissions (1973)

- Berkeley's graduate admissions showed an aggregate gap:
 - **44%** of male applicants admitted vs. **35%** of female applicants
- This looked like evidence of bias against women applicants
- Broken down **by department**,
 - most departments showed **no bias** — some even favored women slightly
- Explanation:
 - women applied in greater numbers to more competitive departments
 - with **lower overall admit rates**

“ *Source: Bickel, Hammel & O'Connell (1975), Science — see References* ”

Lesson 1: Aggregates Can Deceive

- A single overall number can **completely reverse** the story told by its subgroups
- This isn't rare — it shows up in medicine, admissions, sports, sales, and A/B tests
- **Golden Rule:** an aggregate statistic is a starting question, not a final answer

Lesson 2: Always Look for Subgroups

- Before trusting an aggregate comparison, ask:
 - What natural subgroups exist in this data?
 - Are those subgroups distributed **unevenly** across the things being compared?
 - Does splitting by subgroup change — or reverse — the conclusion?
- **Best Practice:** stratify first, aggregate second

Lesson 3: Weighted Averages, Not Simple Averages

- An overall rate is a **weighted average** of subgroup rates, weighted by group size
- A group with a **large, easy** subset can post a high overall number without being the best performer
- A group with a **large, hard** subset can post a low overall number despite being the best performer
- Always ask:
 - what are the weights,
 - and are they the same across what I'm comparing?

Lesson 4: When Aggregation Is (and Isn't) Valid

- **Aggregation is valid when:**
 - Subgroup composition is similar across the groups being compared
 - Or subgroup effects point in the same direction and are of similar size
- **Aggregation is misleading when:**
 - Groups differ heavily in subgroup composition (like caseload mix here)
 - A lurking variable correlates with both the outcome and the grouping
- **When in doubt:** report both the aggregate and the stratified breakdown

Best Practices Checklist

1. Identify plausible confounding or lurking variables
2. Stratify the data before trusting an aggregate comparison
3. Compute both aggregate and subgroup rates
4. Check subgroup weights — are they even across groups?
5. Investigate any reversal between subgroup and aggregate trends
6. Report risk-adjusted or stratified metrics alongside raw ones
7. Stay skeptical of any single summary number

Common Mistakes to Avoid

- Ranking people, teams, or hospitals on raw aggregate rates alone
- Assuming a group difference implies causation without checking confounders
- Ignoring uneven sample sizes or uneven subgroup composition
- Treating a subgroup reversal as a data error instead of investigating it
- Reporting one blended metric when stakeholders need the stratified view

Key Takeaway

- The aggregate number is never the whole story
- Always ask what's hiding underneath the average:
 - a lurking variable,
 - an uneven mix,
 - a hidden group

“ Sometimes the better choice looks worse on paper ”

References

Topic	Source
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